Technical Appendix

Measuring Causal Mental Models at Scale:
An LLM Pipeline for Extracting Directed Causal Narratives from Text

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This appendix documents the instrument described in the Stage-1 proposal in enough detail to evaluate and reproduce it. Section 2 describes the pipeline; Sections 3–5 the taxonomies and prompts; Section 6 the validation design and metrics; Section 7 the benchmark results; Section 8 the failure-mode analyses; Section 9 the full-sample deployment; Section 10 the accompanying repository. Survey responses, human codings and model outputs are not publicly released; the repository contains synthetic examples on which every script runs.

1 Existing methods for measuring mental models

Table 1 compares the instrument with the methods used so far to measure mental models in organizational research and economics.

2 Pipeline architecture

2.1 Overview

The instrument codes open-ended survey responses in two independent stages, each implemented as a separate command-line program calling the OpenAI API.

Stage 1 — cause narratives. Input: the respondent's answer to *"What do you think causes climate change?"* Output: a directed acyclic graph (DAG) whose nodes are factors from a 25-factor taxonomy (Section 3) and whose unique sink is the outcome node *Climate Change* (ID 999).

Stage 2 — policy narratives. Input: the answer to *"What do you think your country's government should do about climate change?"* Output: a vector of seven binary codes A1–A7 (Section 5). Stage 2 is self-contained: it does not read Stage-1 output.

2.2 Model call

Each response is coded by one call to a reasoning model. The live configuration is `o4-mini` with `reasoning_effort = medium`. The system prompt is a fixed text file into which the taxonomy is injected verbatim at a placeholder; the user message contains only the respondent's text. Output is requested as strict JSON with no surrounding prose. The per-call output budget is 4,000 tokens, raised to 10,000 on one automatic retry if the first response is truncated or fails to parse. Calls are

Table 1: Methods for measuring mental models

Method (examples)	Content supplied by	Structure captured	Cost and scale	Main limitation
Likert questionnaires, scenario ratings (Levesque et al. 2001; Webber et al. 2000)	Researcher (items)	None: content and agreement only	Low; scalable	Not a structural measure; unrelated to team process ($\rho \approx -.05$)
Paired-comparison relatedness ratings + Pathfinder / MDS / UCINET (Mathieu et al. 2000; Lim & Klein 2006)	Researcher (fixed concept list)	Undirected network; meaning of "related" (causal, co-occurrence) unspecified	$n(n-1)/2$ judgements (14 concepts = 91); fatigue; mostly lab	Concepts imposed; no causal direction
Concept mapping (Marks et al. 2000; Ellis 2006)	Researcher (concepts, often the structure)	Pre-specified, often sequential or hierarchical	Manual scoring; laborious for large maps	Structure may not exist in respondents' minds
Card sorting (Smith-Jentsch et al. 2001, 2008)	Researcher (concepts); respondent groups them	Clusters; no direction	Tedious with many concepts; cumbersome scoring	Concepts imposed; groupings are not causal links
Repertory grids (Reger & Huff 1993; Hodgkinson 1997; Wright 2004)	Respondent (personal constructs)	Dimensional or hierarchical; not causal	Interview-based	Hard to compare across respondents
Causal / cognitive mapping from interviews or documents (Axelrod 1976; Eden 1992; Markóczy & Goldberg 1995; Hodgkinson et al. 1999)	Respondent, or a pool of researcher constructs	Directed cause–effect graph	Manual elicitation and coding; small samples	Coder judgement hard to reproduce; single-language
Hand-coded text: qualitative coding and DAGs from open-ended surveys (Carley 1997; Kaplan 2008; Andre et al. 2024)	Respondent's own words	Concepts and links coded by researchers; directed if coded so	Trained coders; double coding on subsamples (e.g., 200 responses)	Heavy coder influence; slow; agreement 88% factors, 77% links (Andre et al.)
Computational text analysis: topic models, co-occurrence, embeddings, agent–verb–patient extraction (Hannigan et al. 2019; Ash et al. 2021; Lange et al. 2022)	Corpus text	Associations or event triples; causal direction rarely identified	Fully automated; scalable	Association is not causation; not benchmarked against human coding of structure
This proposal: taxonomy-constrained LLM extraction	**Respondent's own words**, any language; nodes mapped to an *a priori* taxonomy	**Directed causal DAG**; only links the respondent states	**About \$0.005 per response**; 2,964 responses in 12 countries coded	Needs a domain taxonomy; explicit content only; human benchmark so far EN and FR

Notes: Correlations from DeChurch & Mesmer-Magnus (2010); technique characteristics from Mohammed, Ferzandi & Hamilton (2010, Table 2), Lim & Klein (2006) and Becker et al. (2025). For the proposed method, three-way agreement (LLM and two trained coders, $N = 30$ per language) is Fleiss' $\kappa = .64$–$.65$ on edges and .73–.77 on nodes in English and French, with LLM–human agreement at or above human–human agreement; each coding run is tagged with a prompt-and-taxonomy version hash.

stateless: no information from any other response enters the context, so the instrument cannot learn a distribution from earlier items and impose it on later ones. Responses may be in any language; the prompt instructs the model to code on meaning regardless of language.

2.3 Versioning and de-duplication

Every annotation is stored in a cumulative registry keyed by respondent under a `coding_version`: a 12-character SHA-256 prefix of the full system prompt (taxonomy included) concatenated with the model identifier. A run skips any respondent already present under the current hash with a non-null annotation; a change to the prompt, the taxonomy, or the model therefore invalidates prior results rather than silently reusing them. A `-force` flag re-codes everything; a respondent whose stored annotation is null (a failed call) is always retried. Prompt files are under version control, so any `coding_version` can be traced to an exact prompt text.

2.4 Batch mode and cost

Runs can be submitted through the OpenAI Batch API (asynchronous, 50% price reduction). On submission the exact list of items is written to a manifest; resumption reloads from the manifest rather than recomputing from the input, so results cannot be misaligned if the registry changes while a batch is in flight. Based on measured token usage, coding the full sample of 1,729 cause narratives costs approximately $9–31 depending on model and pricing tier.

2.5 Tooling

- `narrative_gpt.py`, `policy_gpt.py`: command-line pipelines. Flags: `-input`, `-output`, `-batch`, `-resume`, `-force`, and `-export-registry`.
- `streamlit_app.py`: web interface covering both stages.
- `annotation_app/app.R`: R Shiny application in which human annotators code the same items under the same taxonomies; cause DAGs are built edge by edge with a factor glossary; policy codes are seven checkboxes with boundary rules. Output format is identical to the model's, so human and model codings are directly comparable.
- `Ancillary/benchmark_agreement.py`: computes all agreement statistics in Section 6.

3 Causal factor taxonomy

The taxonomy was written before data collection from pilot responses, the climate-science literature, technical reports, and press coverage, and refined in pilot coding sessions. Each factor has a definition and example statements (full text in the repository). There is no factor #23.

ID	Factor	Scope note
Natural causes		
1	Climate cycles	Glacial/interglacial rhythms
2	Solar activity & orbital changes	Solar radiation, Milankovitch cycles
3	Volcanic activity	
4	Ocean cycles	El Niño/La Niña, AMOC

ID	Factor	Scope note
5	Geological & tectonic movements	
Human-caused (I): structural and systemic		
6	Greenhouse gases	CO_2, methane, heat-trapping gases; excludes generic "pollution"
7	Overproduction & output pressure	Profit/output maximisation at sector or firm level; supply chains; data centres
8	Energy production & fossil-fuel dependency	Supply side only: extraction, power plants, refineries, grids
9	Industries & industrial processes	Factories, cement, steel, fertilisers, industrial waste
10	Overexploitation of ecosystems & land use	Deforestation, mining, urbanisation, loss of sinks
11	Agriculture & livestock	Methane, fertilisers, pesticides, land conversion
12	Military emissions & warfare	
Human-caused (II): personal and cumulative		
13	Consumerism & overconsumption	Demand for goods, fast fashion, packaging
14	Dietary choices	Meat, dairy, imported food
15	Household energy choices	Heating, insulation, appliances
16	Transportation choices	Private cars, flying, non-EV purchase
17	Waste production & management	Individual waste, recycling, food waste
Political and institutional responsibility		
18	Governmental inaction	Weak or unenforced regulation, failed agreements
19	Lack of low-impact transport systems	Public-investment failure (contrast #16)
20	Corporate influence & lobbying	Corporations as emitters and political actors; greenwashing
21	Economic growth	Capitalism, GDP logic, industrialisation as a system, globalisation; includes population growth
22	Inequality & environmental injustice	
Skepticism, misinformation, alternative theories		
24	Scientific weaponisation	Engineered climate, chemtrails, HAARP
25	Government control	Climate change as pretext for control
26	Data manipulation & media complicity	
999	*Climate change*	Outcome node; may only be a target

4 Stage 1: cause-narrative prompt

4.1 Task structure

The system prompt casts the model as an expert in climate change and causal reasoning performing a classification task, and specifies three ordered tasks:

1. **Node identification.** Identify which taxonomy factors the text mentions, using taxonomy IDs and labels only. If none is identifiable, return an empty graph `{"nodes": [], "edges": []}`.
2. **Edge assessment.** Define directed links among the identified nodes as if drawing a DAG. Climate Change (999) is always the final node and can never be a source. No nodes may be added at this step. Links must be coherent with the text; links not stated in the text are not to be inferred.
3. **Coherence check.** Verify that 999 is present and terminal, and correct any incoherence in place before output.

Output is a JSON object with a `nodes` array (`id`, `label`) and an `edges` array (`source`, `target`).

4.2 Annotation guidelines

Eleven rules resolve recurrent ambiguities. They were written from pilot and benchmark disagreements; rules (j) and (k) were added after the first benchmark and re-validated (Section 8).

(a) Pure lists. Factors listed without stated links each cause 999 directly.

(b) Greenhouse gases (#6). Use #6 for explicit mention of greenhouse gases, CO_2, methane, heat-trapping gases, or atmospheric pollution; not for generic "pollution." Insert #6 as an intermediate node *only* when the text names the gas mechanism ("releases CO_2"); otherwise code the direct edge. This is the parsimony rule.

(c) Energy (#8) vs. sectors. #8 only for supply-side energy systems. Fossil fuels used by a named sector or individual go to that sector (#16 vehicles, #9 factories, #15 households).

(d) Individual vs. systemic. Individual waste #17 vs. industrial waste #9; overproduction #7 vs. economic growth #21; individual transport #16 vs. missing transport infrastructure #19.

(e) Only the respondent's own causes. Do not code causes the respondent attributes to other people's beliefs, or historical events not presented as current causes.

(f) Parenthetical examples count. Factors cited as examples inside parentheses are coded.

(g) Pure descriptions. Effects-only or belief-only text returns an empty graph.

(h) Inaction caused by skepticism. Code the chain skepticism factor $\rightarrow$ 18 $\rightarrow$ 999, not a single node.

(i) "Naturally" as "obviously" does not trigger natural causes.

(j) "Industrialisation" refers to the system: #21, not #9 or #7.

(k) "Profit maximisation": #21 if the economic system is meant, #7 if a sector or firm.

5 Stage 2: policy-narrative coding

5.1 Scheme

Each response receives seven binary codes describing the type of government action proposed—the respondent's *reaction function*.

Code	Label	Mechanism
A1	Restrict / Regulate	Legal compulsion or prohibition (ban, fine, quota, standard)
A2	Price / Incentivise	Price signal (carbon tax, subsidy, rebate, removing fossil subsidies)
A3	Invest / Build	Government as direct provider or funder (infrastructure, R&D, tree planting)
A4	Inform / Educate	Information or communication as the lever (campaigns, curricula)
A5	Adapt / Protect	Reducing vulnerability to impacts (sea walls, resilient crops)
A6	International / Coordinate	Supranational action proposed as part of the response
A7	Skeptical / No action	Government should not act (natural, hoax, too late, markets will solve)

A1–A6 may co-occur freely. A7 is mutually exclusive with A1–A4 and A6; the only valid co-occurrence is A5 (adapt while rejecting mitigation). An all-zero vector is valid.

5.2 Presence threshold and boundaries

A code is 1 if the text contains an explicit noun phrase naming the instrument, a verb phrase describing the action, or a concrete example implying it. It is 0 if the action type is absent, appears only as a sentence-final coda, is too generic to identify an instrument ("do something"), or describes a hoped-for consequence rather than a proposal. The default under uncertainty is 0. Boundary rules separate adjacent codes (e.g., fine enforcing a standard → A1; tax changing a price → A2; subsidy to a private actor → A2; government building directly → A3). **A4 threshold** (added after the first policy benchmark, Section 8): A4 requires an actual information mechanism to be named or strongly implied; bare encouragement verbs ("promote," "encourage," "support"; French *soutenir*, *inciter*, *favoriser*) do not qualify. **A6 threshold**: requires a proposal of international action, not an acknowledgement that the problem is global. The prompt includes a table of language-specific signal expressions (French, Vietnamese, Turkish), eleven worked examples with expected JSON, and a pre-output checklist. Output per response is a JSON record with the seven codes and a $\leq$15-word rationale.

6 Validation design and metrics

6.1 Benchmark sample and coders

Two independent benchmark samples of $N = 30$ responses each: English responses from an online panel (Prolific, United States/United Kingdom) and French responses from a laboratory session.

Two trained human coders (AG, DC; both economists working on climate policy) coded every item in the Shiny application, independently, without communication and without access to model output. The model coded the same items. The same 60 items were used for the Stage-1 (cause) and Stage-2 (policy) benchmarks.

6.2 Units of analysis

Node level (Stage 1). For each respondent, a binary vector of length 25 (one entry per factor). The universe is fixed, so the measure does not depend on graph structure.

Edge level (Stage 1). For each respondent, the universe is the set of all directed pairs among the nodes mentioned by *any* rater (plus 999 as target); each potential edge is scored present/absent per rater. This is the primary measure of agreement on causal structure.

Code level (Stage 2). For each respondent and code A1–A7, present/absent; pooled across codes for a summary.

6.3 Statistics

For each rater pair: observed agreement P_o; Cohen's $\kappa = (P_o - P_e)/(1 - P_e)$; positive percent agreement PPA $= 2n_{11}/(2n_{11} + n_{10} + n_{01})$ (the Dice coefficient, insensitive to shared negatives); negative percent agreement NPA $= 2n_{00}/(2n_{00} + n_{10} + n_{01})$. For three-way agreement, Fleiss'

$$\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}, \qquad \bar{P} = \frac{1}{N}\sum_{i=1}^{N} \frac{\sum_j n_{ij}(n_{ij} - 1)}{k(k-1)}, \qquad \bar{P}_e = \sum_j p_j^2,$$

with N items, $k = 3$ raters, n_{ij} raters assigning category j to item i, and p_j the marginal share of category j. Landis–Koch conventions: 0.61–0.80 substantial, above 0.80 almost perfect. Because graphs are sparse, P_o and NPA are mechanically high; PPA is the demanding measure at the edge level. For low-prevalence codes, κ can be near zero at near-perfect raw agreement (the prevalence paradox); PPA/NPA and prevalence-adjusted κ must be read alongside.

7 Benchmark results

7.1 Stage 1: cause narratives (live configuration)

Table 3: Pairwise agreement, cause narratives. Model = `o4-mini`, live prompt.

		English ($N = 30$)				French ($N = 30$)			
Level	Pair	P_o	κ	PPA	NPA	P_o	κ	PPA	NPA
Edge	Model vs. AG	0.902	0.642	0.699	0.942	0.909	0.640	0.693	0.946
	Model vs. DC	0.902	0.637	0.696	0.942	0.922	0.683	0.728	0.955
	AG vs. DC	0.895	0.627	0.689	0.937	0.907	0.598	0.652	0.946
Node	Model vs. AG	0.956	0.745	0.769	0.976	0.948	0.743	0.772	0.971
	Model vs. DC	0.961	0.798	0.820	0.978	0.949	0.762	0.791	0.971
	AG vs. DC	0.957	0.761	0.784	0.976	0.935	0.686	0.723	0.963

Model–human κ is at or above human–human κ on every row in both languages. Node-level agreement exceeds edge-level agreement throughout: raters agree on which factors are present and

Table 4: Three-way Fleiss' κ_F, cause narratives (model, AG, DC)

	English	French
Edge level ($N_{\text{items}} = 532; 760$)	0.644	0.650
Node level ($N_{\text{items}} = 750; 750$)	0.769	0.731

disagree mainly on structure. Inspection of the most disputed edges shows two structural sources: whether to route a factor through #6 (e.g., 8 → 6 vs. 8 → 999), on which the model is more consistent than the humans, and substitution among #7, #9, and #21 for general references to industry, capitalism, or production. Rules (b), (d), (j), and (k) address both. One taxonomy gap emerged (population growth), folded into #21.

7.2 Stage 2: policy narratives

Table 5: Three-way Fleiss' κ_F by policy code, before and after the A4 threshold rule

	English		French	
Code	Baseline	A4 rule	Baseline	A4 rule
A1 Restrict/Regulate	0.777	0.822	0.686	0.683
A2 Price/Incentivise	0.803	0.850	0.754	0.896
A3 Invest/Build	0.713	0.758	0.548	0.548
A4 Inform/Educate	0.379	0.588	0.793	0.838
A5 Adapt/Protect	0.726	0.726	$-0.02^{\dagger}$	$-0.01^{\dagger}$
A6 International	0.788	0.788	1.000	1.000
A7 Skeptical/No action	0.738	0.738	$0.00^{\dagger}$	$0.00^{\dagger}$
Pooled, all codes ($N_{\text{items}} = 210$)	0.793	**0.840**	0.757	**0.797**

†Raw three-way agreement 96–100%; near-zero κ is the prevalence paradox (almost no French respondent triggered A5 or A7).

7.3 Model comparison

Under an identical prompt, a second frontier reasoning model (`gpt-5.4`) reaches agreement comparable to or higher than `o4-mini`; a smaller variant (`gpt-5.4-mini`) is lower, especially in French, and requires a much larger output budget because of heavier hidden reasoning-token use. Table 7 reports the comparison in the revised-prompt condition in which all three were run.

OpenAI withdrew `o4-mini` from its API on 23 October 2026. We re-ran the benchmark with the live prompts and medium reasoning effort on candidate replacements (Table 6). Agreement is stable across models: on every measure, three of the four candidates are within 0.04 of `o4-mini`, a difference of the size one expects from 30 items per language. The smallest model, `gpt-5-mini`, names the same causes but agrees less on the links between them (0.07–0.09 lower): coding causal structure, not factors, is what demands model capacity. Because each run is tied to its `coding_version` hash, a model change is recorded rather than silent.

Table 6: Three-way Fleiss' κ_F (model, AG, DC) across models, live prompts

	English			French		
	Cause links	Causes	Policy codes	Cause links	Causes	Policy codes
`o4-mini` (June 2026)	0.64	0.77	0.84	0.65	0.73	0.80
`gpt-6.1-terra`	0.66	0.80	0.82	0.62	0.75	0.81
`gpt-5.4-mini`	0.64	0.79	0.80	0.62	0.73	0.77
`gpt-5.4`	0.64	0.78	0.82	0.64	0.75	0.79
`gpt-5-mini`	0.57	0.77	0.81	0.56	0.73	0.76

`o4-mini`: June 2026 validation (Tables above). Cause links: edge level; causes: node level; policy codes: pooled over A1–A7.

8 Failure-mode analyses

8.1 Prompt revision that reduced agreement

After the first benchmark we revised the prompt to (i) add a fourth task scoring each narrative's ambiguity against a criteria list with a free-text note, (ii) add a rule defaulting generic fossil-fuel mentions to #8, and (iii) add rules (j) and (k). Re-benchmarking with the same items and coders showed a reproducible drop (Table 7, rows 1 vs. 2). In French, 10 of 30 items also hit the 10,000-token output budget under the revised prompt, contributing to the fall. An ablation over prompt components isolated the cause: the detailed ambiguity task and the #8 default rule were each sufficient to degrade agreement; rules (j) and (k) were harmless. The live prompt is the original structure plus (j) and (k), and reproduces the original benchmark in both languages. The ambiguity score was also tested as a predictor of human–human disagreement and found weak and inconsistent, so it was dropped rather than retained as a flag.

Table 7: Ablation over prompt components and models: three-way Fleiss' κ_F (edge / node)

	Condition	English	French
1	Original prompt, `o4-mini` (first benchmark)	0.653 / 0.765	—
2	Revised prompt (ambiguity task + #8 default + j,k), `o4-mini`	0.544 / 0.710	0.414 / 0.512
3	Revised prompt, `gpt-5.4`	0.635 / 0.746	0.519 / 0.615
4	Revised prompt, `gpt-5.4-mini`	0.559 / 0.696	0.486 / 0.562
5	Revised prompt without the ambiguity task	0.567 / 0.695	—
6	Revised prompt without the #8 default	0.557 / 0.704	—
7	Original prompt + (j),(k) + a simplified ambiguity task	0.647 / 0.778	0.649 / 0.737
8	**Original prompt + (j),(k) — live**	**0.644 / 0.769**	**0.650 / 0.731**

8.2 Over-coding in the policy stage

The first policy benchmark (Table in Section 7, "Baseline") showed weak agreement on A4 in English and on A2/A3 in French. Rather than rely on κ alone, every model–human disagreement was pulled with the model's own rationale and the response text. The model over-coded relative to humans (English 20 over- vs. 12 under-codes; French 30 vs. 8), and the over-codes concentrated on one mechanism: bare encouragement verbs ("promote," "support," *soutenir*, *inciter*) were treated as sufficient evidence for A2/A3/A4 where the taxonomy's "too generic" exclusion should have applied. A single explicit A4 threshold rule plus a worked example resolved it; pooled κ rose from 0.793 to 0.840 (EN) and 0.757 to 0.797 (FR), with the remaining weak codes (A4 EN, A3 FR) showing similarly low human–human agreement, i.e., genuinely ambiguous categories rather than model error.

8.3 Metric artefacts

French A5 and A7 show $\kappa_F \approx 0$ or negative at 96–100% raw agreement because almost no respondent triggered them. Reporting κ alone would misrepresent a working instrument as broken; the pipeline's agreement script therefore reports P_o, PPA, NPA alongside κ for every code and level.

9 Full-sample deployment

9.1 Sample

1,729 respondents in six countries: Vietnam 544 (laboratory), France 340 (239 laboratory, 101 online), Turkey 308 (laboratory), United States 234 (online), United Kingdom 201 (online), Australia 102 (online). Data collection in further countries is ongoing; the incremental registry codes only newly added respondents.

9.2 Output integrity

Of 1,729 cause narratives, 1 returned a null annotation (JSON truncation; queued for automatic retry) and 93 (5.4%) returned a valid empty graph. Manual inspection of all 93 texts found none nonsensical or low-effort (all $\geq$50 words, median 63): they describe symptoms and evidence of climate change (melting ice, extreme weather) or cite only generic pollution, which factor #6 explicitly excludes. The empty graph is therefore a substantive category—respondents who conflate effects with causes—not coding noise.

9.3 Causal-chain structure

Every root-to-outcome path in each respondent's DAG was enumerated (root = a node with zero in-degree in that respondent's graph; the five natural-cause factors and the three skepticism factors merged into one node each). Path length is not mutually exclusive across respondents: a narrative may contain both a one-step and a two-step chain.

The edge Greenhouse gases $\rightarrow$ Climate Change appears in 58.9% of narratives, but greenhouse gases are the *root* of a chain in only 13.9%: respondents rarely stop at emissions, but overwhelmingly route through them once they reason one step further back. This distinction between a factor's aggregate edge frequency and its frequency as a root cause is only observable because the instrument returns structure rather than mentions.

Table 8: Share of respondents with at least one chain of a given length

Chain length	Share
1 step (factor → 999)	60%
2 steps	51%
3 steps	4.8%
4 steps	0.4%
5 steps	0.06%

Table 9: Most frequent chains (share of all 1,729 respondents)

Chain	Share
One step	
Overexploitation of ecosystems → CC	21.5%
Industries → CC	16.3%
Greenhouse gases → CC	13.9%
Natural causes (combined) → CC	12.0%
Economic growth → CC	10.6%
Two steps	
Energy production → Greenhouse gases → CC	19.9%
Industries → Greenhouse gases → CC	19.8%
Overexploitation of ecosystems → Greenhouse gases → CC	17.5%
Transportation choices → Greenhouse gases → CC	15.3%
Agriculture & livestock → Greenhouse gases → CC	7.1%

10 Reproducibility and repository

The accompanying repository contains:

- the two pipelines, the web interface, and the human-annotation application;
- the live prompts for both stages, the archived revised prompt, and both taxonomies as plain text;
- the agreement script and the exact commands used to compute every table in this appendix;
- synthetic example data (responses, two human codings and a model output) on which the coder and the agreement script run end to end.

Survey responses, human codings and model outputs are not released. Researchers apply the workflow to their own corpus and benchmark: the coder, the annotation application and the agreement script are the same programs that produced the tables here. Model outputs are non-deterministic across calls, so re-running the model on comparable data reproduces agreement up to sampling variation. An open-weights replication of the benchmark, proposed in the main text, will be added to the same repository under its own `coding_version`.